Physical chemistry cumulative exam Nov. 14, 2014

The Clapeyron equation for phase changes is given by $(dP/dT) = \Delta S/\Delta V$.

1. Explain the meaning of the variables in this equation. Make sure to specify the quantities fully. E.g., simply saying that T is the temperature is insufficient; the temperature *of what*.

2. Calculate the melting temperature of ice under a pressure of 80 bar. Give the result in degrees Celsius. The density of ice near 0 °C is 0.92 g cm^{-3} and that of liquid water is 1.00 g cm^{-3} (both practically independent of the pressure). The molar enthalpy of fusion (melting) of ice is 6.01 kJ/mol. (You should know the melting temperature of ice in K units under 1 bar pressure). 1 bar=10^5 Pa.

Keep the units throughout. Results which are off by orders of magnitude because of mixing up cm^3 and m^3 or bar and Pa will be graded as F, irrespectively of the correctness of the logic.

Hint: Use the fact that the melting temperature does not change dramatically with pressure, and thus even 80 bar pressure can be considered as a small change. Therefore, you can equate the derivative (dp/dT) with the *finite* change $\Delta P/\Delta T$.